

INT202 Complexity of Algorithms

Tutorial 4 Q2

Capacity = 5

k	1	2	3	4
B	4	5	2	8
W	2	2	1	3

Q1. Give a tight asymptotic upper bound (Θ notation) on the solution to the following function. You need to justify your answers.

```
T3 (int n){
    if(n <= 2) return 1;
    else return (T3(floor(sqrt(n)))+1);
}
```

	1	2	3	4	5
(1,2)	k ₁	2	2	2	2
(2,4)	k ₁	2	4	6	6
(2,5)	k ₂	2	5	7	7
(3,8)	k ₂	2	5	8	10

$$T(n) = T(2^n)$$

$$S(m) = S(2^m) \rightarrow T(n) = S(m)$$

Q2. There are 4 items, 1 knapsack, and the capacity of the knapsack is 5.

The benefits B of the 4 items are: 4 5 2 8 respectively, and the weights W of the four items are: 2 2 1 3 respectively. Solve the (0-1) Knapsack problem with dynamic programming. You need to show the completed table with computation results.

Q1; 2. substitution method

$$T(n) = T(n^{\frac{1}{2}}) + 1 = T(n^{\frac{1}{4}}) + 2 = T(n^{\frac{1}{8}}) + 3$$

$$\downarrow$$

$$T(n^{\frac{1}{2}}) = T(n^{\frac{1}{4}}) + 1$$

$\downarrow$

$$n^{\frac{1}{2^k}} = 2$$

$\downarrow$

$$\frac{1}{2^k} = \log_2 2$$

$\downarrow$

$$2^k = \log_2 n$$

$\downarrow$

$$k = \log_2 \log_2 n \rightarrow T(n^{\frac{1}{2^k}}) + k = T(2) + \log_2 \log_2 n = 1 + \log_2 \log_2 n = \Theta$$

$$T_3(n) = 1 \cdot T_3(n^{\frac{1}{2}}) + 1$$

$$\hookrightarrow a = 2^m$$

$$T_3(2^m) = T_3(2^{\frac{m}{2}}) + 1$$

$$\hookrightarrow S(m) = S(\frac{m}{2}) + 1$$

$$a = 1 \quad b = 2, \quad f(n) = 1.$$

$$n^{\log_2 1} = 1 = f(n)$$

$$\Rightarrow S(m) = \Theta(\log m) \rightarrow T(n) = \Theta(\log \log n)$$